

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE –
RAIGAD -402 103
Semester Examination – May - 2019

Branch: Electronics and Telecommunication Engineering **Sem.:- IV**
Subject with Subject Code:- Analog Communication Engineering **Marks: 60**
[BTEXC402]
Date:- 16/05/2019 **Time:- 3 Hr.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

(Marks)

- Q.1. A) State and explain Sampling Theorem. What is Nyquist Criteria? A continuous time signal $x(t) = 7 \sin 2500\pi t + 5 \sin 7500\pi t$ is to be sampled. Calculate the minimum sampling frequency. (06)**
- B) With the help of neat diagram explain a typical radio transmitter. (06)**
- Q.2. A) Explain double sideband suppressed carrier technique of AM. (12)**
Also explain generation of the same using Balanced Modulator.
OR
- B) 1. Derive an expression for power relation in an AM wave. (06)**
Assuming that a 750 W carrier is modulated to the depth of 50 percent. Calculate the total power in modulated wave.
- 2. Explain VSB technique of AM, What are its advantages over other AM Techniques (06)**
- Q.3. Answer any two of the following**
- A) Explain generation of Frequency modulation with the help of basic reactance modulator. (06)**
- B) In an FM system, when the audio frequency (AF) is 750 Hz, and AF voltage is 2.6 V, the deviation is 5.2 kHz. If the AF voltage is now increased to 8V what is new deviation? If the AF voltage is (06)**

further raised to 10V while AF is dropped to 200 Hz, what is the deviation? Find Modulation index in each case.

C) Explain concept of Pre-emphasis and De-emphasis. (06)

Q.4. A) What is superheterodyne principle? Explain operation of superheterodyne receiver with the help of block diagram. Also enlist advantages of superheterodyning. (08)

B) Define: Sensitivity, Selectivity, Fidelity, Image frequency (04)

Q.5. A) Draw and explain the circuit diagram of practical diode detector. How AGC obtained from the detector (06)

B) Draw and explain the circuit diagram of ratio detector. How amplitude limiting can be achieved using ratio detector? What are advantages and disadvantages of ratio detector? (06)

Q.6. Answer any two of the following

A) Define Noise. What are different sources of noise? Explain in detail. (06)

B) Two Resistors $15\text{K}\Omega$ and $60\text{K}\Omega$ are at room temperature, for the bandwidth of 100kHz , calculate thermal noise for following conditions. (a) for each resistor (b) for series connection (c) for parallel connections. (06)

C) Define: Noise Figure, Noise Factor (06)

The signal power and noise power measured at the input of an amplifier are $200\text{ }\mu\text{W}$ and $2\text{ }\mu\text{W}$ respectively. If the signal power at the output is 1.5W and noise power is 40mW , calculate the amplifier noise factor and noise figure

Class: B. Tech. EXTC (Second Year)

Sub: Analog Communication Engineering (BTEXC402)

Date: 28/11/2019

Sem: IV

Time: 3 Hrs.

Max. Marks: 60

Instructions to Candidates:

- 1) Attempt any five questions.
- 1) Illustrate the answers with neat sketches, diagram etc. wherever necessary.
- 2) Necessary data is given in the respective questions. If such data is not given it means its knowledge is a part of examination.
- 3) If some part or parameter is noticed to be missing, appropriate data may be assumed and should be mentioned clearly.

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- Q 1 A Define communication, Draw and explain basic block diagram of communication system. (6M)
- B Define modulation; Explain the need of modulation in detail. (6M)
- 2 A For an AM wave with a peak unmodulated carrier voltage $V_c = 10V_p$, a load resistance $R_L = 10\Omega$, and modulation index = 1, determine (8M)
- a. Powers of the carrier and the upper and lower sidebands
 - b. Total sideband power.
 - c. Total power of the modulated wave
 - d. Draw the power spectrum.
- Repeat steps (a) through (d) for a modulation index $m = 0.5$.
- B Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent, and (b) 50 percent. (4M)
- 3 A Draw the block diagram for an Armstrong indirect FM transmitter and describe its operation. (8M)
- B Compare narrowband and wideband FM. (4M)
- 4 A With the help of neat block diagram explain functioning of a super heterodyne receiver list out significance. (6M)
- B Explain the performance characteristics of receiver. (6M)
- 5 A Explain the linear diode (envelope) detector with detail circuit diagram and characteristics. (6M)
- B With neat circuit diagram and necessary equations, explain the phase difference discriminator ratio detector. (6M)
- 6 A An amplifier has a noise figure of 4 dB, a bandwidth of 500Hz and an input resistance of 500Ω . Calculate the input signal voltage needed to yield an output $SNR = 1$ when the amplifier is connected to a signal source of 50Ω at 290K. (4M)
- B Define noise. Explain the classification of noise in detail. (8M)

Paper End

Course: B. Tech.

Branch : E & T C

Semester : V

Subject Code & Name: BTETC503 - Analog Communication Engineering

Max Marks: 60

Date:11/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
 3. Use of non-programmable scientific calculators is allowed.
 4. Assume suitable data wherever necessary and mention it clearly.
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		(Level/CO)	Marks
Q. 1	Solve / Answer the following. (ANY TWO)		
A)	Define Analog Communication and Explain the elements of communication system	CO 01	6
B)	Why modulation is necessary in communication, and what purpose does it serve?	CO 01	6
C)	Describe the concept of multiplexing and its two main techniques: Time Division Multiplexing (TDM) and Frequency Division Multiplexing (FDM).	CO 01	6
Q.2	Solve / Answer the following. (ANY TWO)		
A)	Explain DSBFC Technique in detail with the help of (i) frequency spectrum (ii) Time domain representation (iii) power relation with carrier wave	CO 02	6
B)	A transmitter radiates 10 KW power with the carrier unmodulated, and 10.5 kW when the carrier is modulated by one sinusoidal signal. Calculate the modulation index. If another modulating signal corresponding to 30% modulation is transmitted simultaneously determine the total radiated power	CO 02	6
C)	Provide the mathematical analysis and expression for Amplitude Modulation.	CO 03	6
Q. 3	Solve / Answer the following. (ANY TWO)		
A)	What is Angle modulation? Derive the mathematical expression for FM with frequency spectrum .	CO 05	6
B)	Discuss the concepts of pre-emphasis and de-emphasis in FM systems and their role in improving signal quality.	CO 05	6
C)	It is required to provide a maximum deviation of 75 kHz for the 88-MHz carrier frequency of a VHP FM. transmitter. A FET is used as a capacitive reactance modulator, and the linear portion of its g-curve lies from 320 μ S (at which $V = -2V$) to 830 μ S (at which $V = -0.5V$). Assuming that R is one-tenth of X_c calculate (a) rms value of required modulating voltage (b) The value of the fixed capacitance and inductance of the oscillator tuned circuit across which the reactance modulator is connected	CO 05	6

Q.4 Solve / Answer the following. (ANY TWO)

- A)** Explain the key performance characteristics of receivers: Sensitivity, Selectivity, and Fidelity. CO 07 6
- B)** Describe the Tuned Radio Frequency (TRF) receiver architecture and its advantages and disadvantages. CO 04 6
- C)** Explain the operation of a Phase Discriminator in FM demodulation, and highlight its advantages and disadvantages. CO 04 6

Q. 5 Solve / Answer the following. (ANY TWO)

- A)** Explain the operation of an Envelope Detector in AM demodulation, and discuss its advantages and limitations. CO 04 6
- B)** Define Noise Figure , Noise Factor, and explain their relationship in evaluating the noise performance of electronic devices. CO 04, 06 6
- C)** An AF. signal $20 \sin (2\pi * 500t)$ is used to amplitude modulate a carrier of $50 \sin (2 \pi * 10^5 * t)$ Calculate: CO 04, 06 6
- I. Modulation index
 - II. Sideband frequencies
 - III. Amplitude of each sideband frequency
 - IV. Bandwidth required
 - V. Total power delivered into a load of 600Ω

***** End *****